
Where Does Platonic Convergence Live? A Linear Readout Gains Where the Galaxy Embedding Bends Toward the Label and Loses Where It Bends Away

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Abstract

1 The Platonic Universe tested the Platonic Representation Hypothesis (PRH) by
2 measuring the convergence of foundation-model representations of galaxy images
3 with mutual k -nearest-neighbour agreement (MKNN); we ask whether that con-
4 vergence concentrates where the embedding manifold is flat. A pre-registered
5 raw-decoder mean-curvature statistic is negatively associated with cross-survey
6 agreement, but density removes most of the signal and the one residual case dis-
7 appears once its radial component is removed. A second-order expansion of a
8 linear map restricted to the manifold explains why: a readout sees $\langle w_N, \Pi \rangle$, the
9 second fundamental form contracted with its normal component; under first-order
10 tangent matching an alignment’s second-order error decomposes into a connection
11 mismatch and a relative extrinsic term; mean curvature is only the metric trace of
12 Π and does not in general determine either. On 86,471 labelled galaxies a linear
13 readout gains accuracy where the manifold bends toward the label’s own curvature
14 and loses it where the manifold bends away. In a counterfactual local second-order
15 surrogate of the readout, the manifold’s bending in the readout’s normal direction
16 raises local R^2 at 79–99% of anchors, its sign reversal lowers it at 97–100%, and
17 a random orientation of matched strength shows no such asymmetry and is typi-
18 cally harmful under either sign, on five encoders; across anchors, the mismatch
19 between the label Hessian and the probe-facing Hessian is negative wherever the
20 label has local structure and survives cross-fitting and a change of encoder. Mean
21 curvature of a single representation does not explain weak Platonic convergence;
22 the obstruction is relative and, for the fitted global linear alignment, present at first
23 order.

24 1 Introduction

25 The Platonic Representation Hypothesis [Huh et al., 2024] holds that models trained on different data
26 converge toward a shared representation; Duraphe et al. [Duraphe et al., 2025] tested it on galaxies
27 imaged by two surveys (Legacy Survey and Hyper Suprime-Cam, HSC), which share only 0.3 to 2.3
28 percent of nearest neighbours across surveys under a family of vision models. If two models disagree
29 most where the embedding surface bends sharply, local mean curvature should associate negatively
30 with local cross-survey agreement.

31 We contribute (i) a pointwise mean-curvature and second-fundamental-form instrument from a
32 trained decoder, validated against analytic truth; (ii) a density-controlled test of the curvature-
33 alignment prediction, which finds no density-free effect; (iii) the residual expansion of a linear map
34 on a submanifold; and (iv) known-surface, cross-encoder and interventional tests: a linear readout

gains local accuracy where the manifold bends toward the label’s curvature and loses it where it bends away, at a large majority of anchors on five encoders.

2 Pointwise curvature from a decoder

Data. DINOv3 [Siméoni et al., 2025] ViT-B/16 embeddings of 101,725 galaxies imaged by both HSC [Aihara et al., 2018] and the Legacy Survey [Dey et al., 2019] (Platonic Universe release [Dupaphe et al., 2025]); we draw 10,000 rows once with a recorded seed. A second set of 86,471 single-survey galaxies embedded by a supervised ImageNet-21k ViT-B/16 (“ViT-B” below; not DINOv3) with catalogue labels (r -band magnitude, photometric redshift, smooth fraction, stellar mass; the As- troPT galaxy set [Smith et al., 2024]) serves the readout experiment. We unit-normalize embeddings (data on S^{767}).

Instrument. Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^{768}$ be the decoder of a plain auto-encoder (three hidden layers of width 250, SiLU), fit on 80% of rows; we evaluate curvature on held-out rows. With $J = DF(z)$, metric $g = J^\top J$, normal projector $P_N = I - Jg^{-1}J^\top$ and second fundamental form $\Pi = P_N D^2 F(z)$, the mean-curvature vector of the raw decoder surface is $H_{\text{raw}} = \text{tr}_g \Pi$ (un-normalized trace; some authors divide by d), by automatic differentiation (cf. [Lee and Park, 2023, Acosta et al., 2023]), at points where DF has full column rank d (likewise $D(F/\|F\|)$ after projection). On the unit sphere the Euclidean mean-curvature vector of every d -submanifold has the universal radial component $-d\hat{x}$, which says nothing about shape; differentiating $F/\|F\|$ instead splits H exactly as $-d\hat{x} + H^S$ ($H_{\text{rad}} = -d$ to 10^{-14}), and H^S carries the non-radial mean-curvature component. Section 3 uses the pre-registered $\|H_{\text{raw}}\|$; Sections 4–5 use the projected decoder.

Validation. Against closed-form curvature, at $D = 768$ an in-sphere surface $G(z) = Q \text{ normalize}([\phi(z); 0.8h(z); 0 \dots])$ (ϕ inverse stereographic $\mathbb{R}^{16} \rightarrow S^{16}$, h four Gaussian bumps, Q a rotation) gives rank 0.94 against the true $\|H^S\|$, direction cosine 0.999 and magnitude ratio 1.00 for the full vector.

3 Curvature and cross-survey convergence

We correlate the per-point mutual k -nearest-neighbour score $\text{MKNN}_i = |N_k(z_1)_i \cap N_k(z_2)_i|/k$ at $k = 20$ between the HSC and Legacy Survey embeddings with $\|H_{\text{raw}}\|$ at $d \in \{20, 25, 32\}$. The raw association $\rho(\|H_{\text{raw}}\|, \text{MKNN})$ is -0.112 , -0.128 and -0.024 at $d = 20, 25, 32$ ($p < 10^{-5}$, $< 10^{-5}$, 0.009). But k -NN statistics are density-sensitive [Radovanović et al., 2010]: local density ($k = 30$ estimate on the Legacy embedding) correlates with agreement at -0.21 at every d and with $\|H_{\text{raw}}\|$ at $+0.43$, $+0.32$ and $+0.01$, so residualizing on density leaves -0.024 , -0.066 and -0.022 . Under a permutation null within density strata only $d = 25$ survives ($p < 5 \times 10^{-5}$), and subtracting the radial projection of H_{raw} (a projection of H_{raw} , not a recomputation on $F/\|F\|$) collapses it to -0.023 .

4 What a linear map sees on a curved manifold

Let $M \subset \mathbb{R}^D$ be a d -dimensional submanifold, $d < D$; at x_0 take an orthonormal tangent frame J , the second fundamental form Π (normal-valued, symmetric $d \times d$), $H = \text{tr} \Pi$, and for a vector w its tangent and normal parts $w_T + w_N$. In normal coordinates u at x_0 , M is $x_0 + Ju + \frac{1}{2}\Pi(u, u) + O(|u|^3)$ [Lee, 2018], so

$$[\nabla_M(w \cdot x)]_J = J^\top w, \quad \text{Hess}_M(w \cdot x) = \langle w_N, \Pi \rangle, \quad \Delta_M(w \cdot x) = \langle w, H \rangle. \quad (1)$$

For $M \subset S^{D-1}$, $\Pi = \Pi^S - g\hat{x}$ splits off the sphere’s radial direction, so $\text{Hess}_M(w \cdot x) = \langle w_S, \Pi^S \rangle - (w\hat{x})g(2)$, with w_S the part of w_N orthogonal to $\hat{x}$: a linear readout of a unit-normalized embedding is bent by the manifold’s shape and, whatever the manifold, by the sphere in proportion to the readout’s own value $\hat{y} - b_0$. For a label y on M , a global affine probe $\hat{y} = w \cdot x + b_0$ and a patch $u \sim N(0, s^2 I_d)$ in normal coordinates, embedding and label locally C^4 , with $\bar{c}$ the patch-mean residual, $\delta = [\nabla_M y]_J - J^\top w$, $\Delta = \text{Hess}_M y - \langle w_N, \Pi \rangle$ (all tensor norms and cosines use

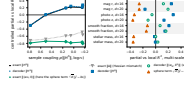


Figure 1: (a) Known surface, linear label, exact geometry: three-control partial vs local R^2 as the density-curvature coupling is set by construction (γ from -1 to 1). (b) Galaxies, 512 anchors, $k = 2,048$, multi-scale control: the decoder’s $\|H^S\|$ (filled squares), shape term (open circles), sphere term (open triangles).

label	$\ H^S\ $		shape $\ \langle w_S, \Pi^S \rangle\ $		sphere $\sqrt{d} \hat{y} - b_0 $		mismatch $\ \Delta\ _g$		alignment	
	$d=16$	$d=20$	16	20	16	20	16	20	16	20
mag_r	+0.25	-0.17	+0.12	+0.19	-0.26	-0.26	-0.39	-0.40	+0.35	+0.43
photo_z	+0.28	+0.13	-0.06*	-0.12	-0.33	-0.36	-0.45	-0.44	+0.26	+0.24
smooth_fraction	+0.21	+0.15	+0.07*	+0.16	-0.13	-0.15	-0.09*	-0.17	+0.19	+0.07*
stellar_mass	-0.02*	-0.00*	+0.00*	+0.03*	-0.35	-0.35	-0.04*	-0.09*	+0.01*	+0.02*

Table 1: Galaxies, 512 anchors, $k = 2,048$: rank-partial Spearman against local R^2 under the multi-scale density control. * not significant at 0.05.

$$\langle A, B \rangle_g = g^{ia} g^{jb} A_{ij} B_{ab},$$

$$\mathbb{E}_{\text{patch}}[(y - \hat{y})^2] = \bar{c}^2 + s^2 \|\delta\|^2 + \frac{1}{2} s^4 \|\Delta\|_g^2 + O(s^4 \|\delta\|) + O(s^6). \quad (3)$$

For two embeddings F, G of the same objects in a common chart u and a linear map A with $AJ_F = J_G$, the same expansion gives $G - (G_0 + A(F - F_0)) = \frac{1}{2} [J_G(\Gamma^G - \Gamma^F) + \Pi_G - A\Pi_F](u, u) + O(|u|^3)$, with Γ the induced connections: the second-order residual contains a connection mismatch and a *relative* extrinsic term, whose target-normal component is $\Pi_G - P_N^G A \Pi_F$.

So $\|H\|$ need not correlate with either term; and since the second-order term scales as s^4 with s set by density at fixed k , patch-scored curvature effects are density-sensitive.

5 Probe-facing curvature: bending toward the label helps, bending away hurts

Statistic: five-fold out-of-fold ridge probe [Alain and Bengio, 2016] at the pre-registered $\alpha = 100$ from embedding to label; local R^2 inside each of 512 anchors’ 2,048-neighbourhoods; rank-partial Spearman against local R^2 with Freedman–Lane p -values [Freedman and Lane, 1983], controlling for log patch radius, local label variance and count and, stronger, log radius at five scales.

Known surface. Resampling the validation surface with weight $\|H^S\|^\gamma$ (Figure 1a) moves the $\|H^S\|$ partial from -0.29 to $+0.26$ while $\|\langle w_N, \Pi \rangle\|_g$ holds at -0.74 to -0.80 , dominated by the sphere term of Eq. (2).

Galaxies. Here the two terms of Eq. (2) trade places: the shape term dominates the probe-facing Hessian (rank 0.96–0.99). We estimate $\text{Hess}_M y$ per anchor by a least-squares quadratic of the label on the tangent-projected coordinates $u = g^{-1} J^\top(x - x_0)$ of its 2,048 neighbours (decoder-based Monge coordinates, so the finite-patch coefficient estimates the covariant Hessian). Three regularities follow (Table 1; Appendix A); the sphere term is negative on every label and both d (-0.13 to -0.36) under the sealed probe; with a weak-ridge probe ($\alpha = 1$, Appendix B) it stays negative for magnitude and smooth fraction ($-0.21, -0.12$) and vanishes for redshift and stellar mass. For magnitude, redshift and smooth fraction the mismatch is negative (-0.09 to -0.45 ; unchanged under cross-fitting, Appendix B) and the alignment $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$ positive ($+0.07$ to $+0.43$); stellar mass shows no mismatch or alignment signal here. The shape term alone is label-signed (positive for magnitude and smooth fraction, null to weakly negative for redshift): with $K = \langle w_N, \Pi \rangle$, $\|\Delta\|_g^2 = \|\text{Hess}_M y\|_g^2 - 2\langle \text{Hess}_M y, K \rangle_g + \|K\|_g^2$, so the cross term decides for infinitesimal scaling from zero and $\|K\|_g^2$ is the finite-magnitude overshoot penalty. The same regression on the readout $w \cdot x$ returns the point cloud’s own $\langle w_N, \Pi \rangle$ (median cosine with the decoder’s 0.55–0.65).

Bending toward helps, bending away hurts. Within the quadratic surrogate of Eq. (3), the Hessian-mismatch term is lower than at $K = 0$ iff $2\langle \text{Hess}_M y, K \rangle_g > \|K\|_g^2$: the manifold must bend toward the label’s own curvature, and not too far. (Liu et al. [2023] derive the optimal normal weight of least squares on a hypersurface as a uniqueness condition; manifold capacity [Chung et al., 2016, Slatton et al., 2026], flattening [Psenka et al., 2024], latent-curvature studies [Kaufman and Azencot, 2023] and local-regression bias [Cheng and Wu, 2013] use dimension, radius or magnitude; none conditions on the sign of $\langle \text{Hess}_M y, K \rangle$.) No anchor is flat, so we test the condition with a counterfactual local second-order surrogate of the readout. Writing $K = K_S - (w \cdot \hat{x})_g$ with $K_S = \langle w_S, \Pi^S \rangle$ the in-sphere shape term, each anchor’s 2,048 neighbours are scored by $\hat{y}_t(u) = c + a^\top u + \frac{1}{2} K_{\text{sph}}(u, u) + \frac{t}{2} K_S(u, u)$ with a the readout’s tangent gradient, K_S from the decoder and c refit: $t = 1$ is the manifold’s bending as seen in the readout’s normal direction (“bending toward the label” means $\langle R, K_S \rangle > 0$ for the remaining label curvature $R = \text{Hess}_M y - K_{\text{sph}}$), $t = -1$ the sign reversal of that probe-facing shape quadratic, $t = 0$ shape-flat. In the Gaussian model $t = 1$ beats shape-flat iff $2\langle R, K_S \rangle_g > \|K_S\|_g^2$; on the actual neighbours, with r_0 the centred shape-flat residual and q_c the centred quadratic $\frac{1}{2} K_S(u_i, u_i)$, the exact criterion is $2\langle r_0, q_c \rangle > \|q_c\|^2$ with optimum $t^* = \langle r_0, q_c \rangle / \|q_c\|^2$ (Appendix E). On five encoders and all four labels, $t = 1$ beats shape-flat at 79–99% of anchors and $t = -1$ is worse at 97–100%, while a random orientation with the same centred quadratic amplitude shows no such asymmetry and is typically harmful under either sign ($t = 1$ beats shape-flat at 22–36% of anchors, $t^* \approx 0$; Appendix E). The probe-selected shape quadratic is locally error-reducing; under the second-order model that is the signature of bending toward the remaining label curvature. This within-anchor question differs from Table 1’s cross-anchor one (does variation in mismatch explain variation in local R^2) and can hold where that has no signal, as for stellar mass. t^* sits at 1.5 to 3.6 throughout, consistent with the globally ridge-regularized probe under-using a locally beneficial second-order term.

Across encoders. The quadratic-chart estimator’s mean-bending trace at the same anchors of five encoders (the readout set’s supervised ViT-B, DINOv3 ViT-B/16, CLIP-B, ConvNeXt-B, ViT-L; $d = 16$) has density-controlled associations with the magnitude probe’s local R^2 of both signs (-0.24 to $+0.27$), as Eq. (1) allows. Per encoder (Appendix D) the decoder test is stable where the expansion predicts: the mismatch partial is negative for magnitude and redshift on all five encoders (-0.32 to -0.61 ; cross-fitted -0.28 to -0.62) and the alignment positive for magnitude on all five ($+0.13$ to $+0.53$) and for redshift on four; the shape term’s sign varies with encoder as well as label; for the mismatch and alignment hypotheses stellar mass stays null but for one mismatch cell (-0.12).

6 Discussion

A linear readout of a galaxy embedding gains accuracy where the manifold, seen in the readout’s normal direction, bends toward the label’s curvature and loses it where that bending is reversed: gain at a large majority of anchors and loss at nearly every anchor, on five encoders. The residual expansion names the quantity, and a decoder’s pointwise second fundamental form measures it at production scale, validated against analytic truth for H^S and in substantial agreement with the data. Under first-order tangent matching, the relevant second-order extrinsic normal term of a linear alignment map [Bangachev et al., 2026] is $\Pi_G - P_N^G A \Pi_F$, not one embedding’s mean curvature; the full residual also carries the connection mismatch; our pilot (Appendix C) finds a 44% first-order tangent-frame residual, which leaves its relative-II column only an extrinsic diagnostic. On this survey pair, single-model mean curvature has no robust density-controlled association with cross-survey convergence.

Limitations. One survey pair. The label’s Hessian is a local estimate; its norm is reliably ranked (split-half ρ 0.91–0.94) but its direction only moderately (tensor cosine 0.19–0.35). On a synthetic surface with known curvature the mismatch partial reproduces but the alignment partial has no stable sign; Appendix E carries the alignment claim. Mismatch and alignment are the primary hypotheses, the rest exploratory. Permutation p -values assume anchor exchangeability under overlapping patches.

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208 Appendix A: decoder ablations

209 Table 2 repeats the four quantities of Table 1 at $d = 16$ under the multi-scale density control for two
210 further decoder initialisation seeds and for a decoder of width 400^3 instead of 250^3 (all fits reach
211 variance explained 0.952). Every significant sign of the main table is retained.

label	quantity	seed 0	seed 1	seed 2	400 ³
mag_r	shape	+0.12	+0.05*	+0.12	+0.05*
	sphere	−0.26	−0.26	−0.27	−0.26
	mismatch	−0.39	−0.35	−0.34	−0.34
	alignment	+0.35	+0.31	+0.37	+0.41
photo_z	shape	−0.06*	−0.09*	−0.09	−0.06*
	sphere	−0.33	−0.35	−0.34	−0.35
	mismatch	−0.45	−0.43	−0.42	−0.41
	alignment	+0.26	+0.24	+0.25	+0.25
smooth_fraction	shape	+0.07*	+0.20	+0.15	+0.10
	sphere	−0.13	−0.15	−0.14	−0.11
	mismatch	−0.09*	−0.05*	−0.08*	−0.07*
	alignment	+0.19	+0.14	+0.09	+0.10
stellar_mass	shape	+0.00*	−0.06*	+0.05*	+0.09
	sphere	−0.35	−0.35	−0.35	−0.35
	mismatch	−0.04*	+0.03*	+0.02*	+0.03*
	alignment	+0.01*	−0.02*	+0.03*	−0.02*

Table 2: Decoder ablations, $d = 16$, 512 anchors, multi-scale density control. * not significant at 0.05.

212 Appendix B: sensitivity of the mismatch and sphere columns

213 Cross-fitting: each 2,048-patch is split at random into halves; $\text{Hess}_M y$ is fitted on one half and local
214 R^2 scored on the other (both directions shown), so the Hessian estimate and the outcome share no
215 rows. The split-half cosine between the two Hessian estimates is the reliability of the estimate. Weak
216 ridge: the probe refit at $\alpha = 1$ instead of 100 (global out-of-sample R^2 rises by 0.10 to 0.15), which
217 substantially reduces the shrinkage of extreme predictions; the sphere term is recomputed from that
probe.

label	d	mismatch, cross-fit		alignment, cross-fit		Hess. split cos p50	sphere, $\alpha = 1$ (main: $\alpha = 100$)
		fit A/score B	fit B/score A	fit A/score B	fit B/score A		
mag_r	16	−0.38	−0.34	+0.33	+0.31	+0.35	−0.21 (−0.26)
mag_r	20	−0.43	−0.40	+0.39	+0.37	+0.24	−0.22 (−0.26)
photo_z	16	−0.42	−0.36	+0.23	+0.20	+0.32	+0.05* (−0.33)
photo_z	20	−0.42	−0.42	+0.22	+0.22	+0.20	+0.03* (−0.36)
smooth_fraction	16	−0.10	−0.17	+0.17	+0.19	+0.33	−0.12 (−0.13)
smooth_fraction	20	−0.15	−0.21	+0.05*	+0.03*	+0.23	−0.13 (−0.15)
stellar_mass	16	+0.01*	−0.04*	−0.03*	+0.06*	+0.31	−0.02* (−0.35)
stellar_mass	20	−0.06*	−0.07*	+0.01*	−0.00*	+0.19	−0.02* (−0.35)

Table 3: Cross-fitted mismatch and alignment partials (multi-scale control), split-half reliability of the label Hessian, and the sphere-term partial under a weak-ridge probe.

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219 Appendix C: relative second fundamental form, pilot

220 Two sphere-projected decoders (HSC = F , Legacy = G ; Phase 7 protocol), a global ridge map A
221 from x_F to x_G (fit on the 8,000 training rows), 2,048 seeded anchors. Columns: holdout R^2 of A ;
222 median first-order obstruction $\|AJ_F - J_G L\|/\|AJ_F\|$ with $L = J_G^+ AJ_F$; then multi-scale density-
223 controlled partials against MKNN ($k = 20$; log radius at $k \in \{10, 30, 100, 300\}$ in both spaces)
224 of each decoder’s $\|H^S\|$, the first-order obstruction, $\|\Pi_G(L, L) - P_N^G A \Pi_F\|$ (because first-order
225 matching $AJ_F = J_G$ fails at the 44% level, the second derivative of the alignment residual is not
226 coordinate invariant; in the decoder chart its target-normal quadratic coefficient additionally contains
227 $-P_N^G AJ_F \Gamma^F$, so relative II is reported only as an extrinsic diagnostic), the same with L fitted on 256
228 neighbours’ latent codes, and a decoder-free estimate from the quadratic coefficient of the alignment
229 residual on F ’s tangent coordinates.

run	R_A^2	tan. resid	$\ H^S\ $ HSC	$\ H^S\ $ Legacy	tan. resid	Π_{rel}	Π_{rel} (local L)	Π_{rel} (data)
$d = 20$, seed 0	0.64	0.44	−0.15	+0.04*	−0.18	+0.04*	−0.18	+0.13
$d = 25$, seed 0	0.64	0.43	−0.10	+0.06	−0.16	+0.05	−0.23	+0.13
$d = 20$, seed 1	0.64	0.44	−0.11	+0.05	−0.19	+0.04	−0.20	+0.14

Table 4: Relative-II pilot. * not significant at 0.05.

Appendix D: the same test on four further encoders

Table 5 repeats Table 1 at $d = 16$ on the same 86,471 galaxies embedded by four further encoders from the same release (DINOv3 ViT-B/16, CLIP ViT-B, ConvNeXt-B, ViT-L; widths 768, 512, 1,024, 1,024; each unit-normalized), with a decoder of the same width and protocol fitted per encoder (seed 0; variance explained 0.966, 0.985, 0.965, 0.968 for the four new fits), 512 anchors, $k = 2,048$, multi-scale density control. The first row per label is the probe’s global out-of-sample R^2 ; Table 6 gives the cross-fitted mismatch and alignment and the split-half reliability of the label Hessian per encoder.

label	quantity	ViT-B	DINOv3	CLIP-B	ConvNeXt-B	ViT-L
mag_r	global R^2	0.52	0.57	0.50	0.52	0.52
	shape	+0.12	−0.11	+0.16	+0.16	−0.21
	sphere	−0.26	−0.23	−0.40	−0.24	−0.08*
	mismatch	−0.39	−0.57	−0.34	−0.32	−0.44
	alignment	+0.35	+0.53	+0.32	+0.21	+0.13
photo_z	global R^2	0.51	0.56	0.49	0.49	0.51
	shape	−0.06*	−0.05*	−0.17	+0.29	−0.16
	sphere	−0.33	−0.21	−0.29	−0.46	−0.44
	mismatch	−0.45	−0.61	−0.45	−0.48	−0.46
	alignment	+0.26	+0.33	+0.06*	+0.22	+0.14
smooth_fraction	global R^2	0.53	0.59	0.49	0.56	0.55
	shape	+0.07*	−0.08*	+0.20	+0.24	+0.06*
	sphere	−0.13	−0.06*	+0.05*	−0.15	−0.11
	mismatch	−0.09*	−0.33	−0.25	−0.20	−0.01*
	alignment	+0.19	−0.02*	+0.24	+0.04*	+0.14
stellar_mass	global R^2	0.48	0.54	0.50	0.49	0.49
	shape	+0.00*	−0.15	−0.10	−0.11	−0.13
	sphere	−0.35	−0.26	−0.38	−0.40	−0.38
	mismatch	−0.04*	+0.03*	−0.12	−0.06*	−0.02*
	alignment	+0.01*	−0.05*	−0.01*	+0.07*	−0.03*

Table 5: Cross-encoder replication, $d = 16$, multi-scale density control. * not significant at 0.05.

Appendix E: intervening on the readout’s normal component

The partials of Table 1 compare anchors with one another and cannot say whether reversing the probe-facing shape term hurts relative to removing it, since no anchor is flat. Here we score a counterfactual local second-order surrogate of the readout at a fixed manifold. At each anchor the fitted ridge weight is split into tangent, radial and in-sphere normal parts $w = w_T + w_{\text{rad}} + w_S$; we score the anchor’s 2,048 neighbours by the data-side tangent-plus-sphere readout $(w_T + w_{\text{rad}}) \cdot x$ plus tq with the local intercept refit, where $q = \frac{1}{2}\langle w_S, \Pi^S \rangle(u, u)$ is the decoder’s in-sphere second-order term in the anchor’s chart coordinates: $t = 1$ is the manifold’s bending as seen in the readout’s normal direction, $t = -1$ its sign reversal, $t = 0$ shape-flat (the sphere term $-(w \cdot \hat{x})g$ is kept in the base). Because the intercept is refit for every t , the local sum of squares is exactly $\text{SSE}(t) = \|r_0 - tq_c\|^2$ with r_0 the centred residual of the shape-flat predictor and $q_c = q - \bar{q}\mathbf{1}$ the centred quadratic on the anchor’s neighbours; hence $t^* = \langle r_0, q_c \rangle / \|q_c\|^2$ and $t = 1$ beats shape-flat iff $2\langle r_0, q_c \rangle > \|q_c\|^2$. This is the finite-sample counterpart of the tensor-level condition $2\langle R, K_S \rangle_g > \|K_S\|_g^2$ of Section 5, not the same quantity: the tensor version needs the estimated label Hessian, the empirical one does not. Columns: fraction of anchors where $t = 1$ beats shape-flat (help) and where $t = -1$ is worse than shape-flat (hurt), the median change in local R^2 at $t = \pm 1$, the median t^* ; then the same for a random

encoder	label	mismatch, cross-fit		alignment, cross-fit		Hess. split cos p50
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	
DINOv3	mag_r	−0.60	−0.62	+0.50	+0.50	+0.35
	photo_z	−0.56	−0.59	+0.28	+0.27	+0.24
	smooth_fraction	−0.32	−0.29	−0.03*	−0.02*	+0.31
	stellar_mass	+0.03*	+0.02*	−0.03*	−0.07*	+0.22
CLIP-B	mag_r	−0.28	−0.35	+0.28	+0.29	+0.31
	photo_z	−0.41	−0.40	+0.08*	+0.06*	+0.25
	smooth_fraction	−0.22	−0.24	+0.15	+0.20	+0.28
	stellar_mass	−0.14	−0.13	−0.04*	+0.01*	+0.19
ConvNeXt-B	mag_r	−0.36	−0.34	+0.19	+0.21	+0.29
	photo_z	−0.46	−0.45	+0.18	+0.19	+0.30
	smooth_fraction	−0.15	−0.19	−0.02*	+0.03*	+0.32
	stellar_mass	−0.05*	−0.01*	+0.14	−0.03*	+0.29
ViT-L	mag_r	−0.46	−0.49	+0.09*	+0.15	+0.30
	photo_z	−0.44	−0.42	+0.09*	+0.13	+0.24
	smooth_fraction	−0.03*	−0.04*	+0.11	+0.07*	+0.30
	stellar_mass	−0.06*	+0.01*	−0.04*	−0.03*	+0.21

Table 6: Cross-fitted mismatch and alignment partials per encoder (multi-scale control) and split-half reliability of the label Hessian.

254 in-sphere normal direction v whose quadratic is rescaled to the same centred amplitude on the actual
 255 neighbours, $\|q_{v,c}\| = \|q_c\|$ (matching $\|v\|$ to $\|w_S\|$ instead leaves the contracted tensor at a fraction
 256 of the fitted one, since Π^S spans at most $d(d+1)/2$ of the ~ 750 normal directions; matching
 257 $\|\langle v, \Pi^S \rangle\|_g$ gives the same picture; Supplement 12). The anchors' neighbourhoods overlap (each
 258 point lies in about twelve of them), so the fractions are not counts of independent trials; a maximal
 259 set of anchors whose pairwise overlap is at most 5% of k has 19–21 members per run, on which $t = 1$
 260 beats shape-flat at 81–100% (one-sided sign test $p \leq 4 \times 10^{-3}$ in every cell) and $t = -1$ is worse
 261 at 95–100% ($p \leq 3 \times 10^{-5}$).

run	label	decoder in-sphere shape term					random orientation, matched $\ q_c\ $				
		help	hurt	$\Delta R^2(+1)$	$\Delta R^2(-1)$	t^*	help	hurt	$\Delta R^2(+1)$	$\Delta R^2(-1)$	
ViT-B, $d = 16$	mag_r	0.97	1.00	+0.030	-0.051	2.3	0.26	0.80	-0.009	-0.009	
	photo_z	0.99	0.99	+0.028	-0.038	3.0	0.33	0.68	-0.005	-0.006	
	smooth_fraction	0.90	1.00	+0.021	-0.040	1.7	0.24	0.78	-0.008	-0.010	
	stellar_mass	0.96	0.99	+0.026	-0.040	2.6	0.29	0.70	-0.006	-0.006	
ViT-B, $d = 20$	mag_r	0.99	1.00	+0.031	-0.049	2.6	0.24	0.78	-0.007	-0.008	
	photo_z	0.99	1.00	+0.029	-0.038	3.6	0.36	0.71	-0.004	-0.005	
	smooth_fraction	0.90	1.00	+0.020	-0.035	1.9	0.22	0.77	-0.007	-0.008	
	stellar_mass	0.97	1.00	+0.027	-0.038	3.1	0.32	0.72	-0.005	-0.005	
DINOv3, $d = 16$	mag_r	0.92	0.99	+0.018	-0.031	2.0	0.29	0.71	-0.007	-0.006	
	photo_z	0.96	0.99	+0.020	-0.030	2.6	0.29	0.71	-0.006	-0.004	
	smooth_fraction	0.93	1.00	+0.024	-0.055	1.5	0.22	0.78	-0.014	-0.013	
	stellar_mass	0.95	1.00	+0.016	-0.028	2.1	0.26	0.74	-0.005	-0.005	
CLIP-B, $d = 16$	mag_r	0.95	0.99	+0.018	-0.031	2.2	0.30	0.68	-0.006	-0.005	
	photo_z	0.99	1.00	+0.019	-0.026	3.4	0.36	0.64	-0.003	-0.003	
	smooth_fraction	0.92	0.99	+0.014	-0.027	1.7	0.25	0.69	-0.006	-0.005	
	stellar_mass	0.97	1.00	+0.020	-0.028	2.8	0.35	0.68	-0.004	-0.005	
ConvNeXt-B, $d = 16$	mag_r	0.92	1.00	+0.018	-0.031	2.0	0.26	0.73	-0.007	-0.007	
	photo_z	0.99	1.00	+0.018	-0.026	2.9	0.28	0.64	-0.004	-0.003	
	smooth_fraction	0.88	0.98	+0.016	-0.036	1.6	0.23	0.75	-0.009	-0.008	
	stellar_mass	0.99	1.00	+0.022	-0.032	3.1	0.29	0.65	-0.005	-0.003	
ViT-L, $d = 16$	mag_r	0.96	0.99	+0.022	-0.034	2.6	0.32	0.74	-0.005	-0.006	
	photo_z	0.97	0.99	+0.018	-0.025	3.2	0.34	0.67	-0.004	-0.003	
	smooth_fraction	0.79	0.97	+0.013	-0.029	1.6	0.25	0.77	-0.006	-0.008	
	stellar_mass	0.98	1.00	+0.022	-0.032	2.9	0.30	0.67	-0.004	-0.004	

Table 7: Counterfactual scaling of the in-sphere shape term in a local second-order surrogate of the readout, 512 anchors per run. The manifold’s bending as seen in the readout’s normal direction helps at a large majority of anchors, its sign reversal hurts at nearly every anchor, a random orientation of the same centred quadratic amplitude shows no such asymmetry and is typically harmful under either sign; $t^* > 1$ throughout, consistent with the globally ridge-regularized probe under-using a locally beneficial term.